

In-Class Lab: Pearson Correlation Test Using Real World Advertising Data

Objective:

In this lab, students will explore the relationship between advertising spending and sales revenue. Using the Pearson correlation coefficient, they will determine the strength and direction of the correlation between different advertising platforms (TV, Radio, Newspaper) and sales.

Dataset:

You can download the **Advertising Dataset** from this [link](#). This dataset contains information on the money spent on TV, radio, and newspaper advertising, along with the sales revenue.

Dataset Overview:

TV: Advertising budget for TV in USD

Radio: Advertising budget for radio in USD

Newspaper: Advertising budget for newspaper in USD

Sales: Sales revenue in USD

Steps for the Mini-Project:

Part 1: Data Familiarization

1. **Download the dataset** from [Kaggle](#).

2. **Understand the variables:**

TV: TV advertising spending (independent variable)

Radio: Radio advertising spending (independent variable)

Newspaper: Newspaper advertising spending (independent variable)

Sales: Sales revenue (dependent variable)

3. **Identify research questions:**

Is there a significant relationship between TV advertising and sales?

How does radio advertising correlate with sales?

Does newspaper advertising have a strong correlation with sales?

Part 2: Pearson Correlation Analysis

1. Select Variables:

Choose one pair of variables to analyze first (e.g., **TV Advertising and Sales**).

You can analyze more pairs as additional tasks (e.g., **Radio and Sales**, **Newspaper and Sales**).

2. Manual Calculation (Optional):

You can walk through a manual calculation process as described earlier to strengthen understanding, but the focus should be on software-based analysis.

3. Python-Based Calculation:

Use Python to calculate the Pearson correlation coefficient:

4. Interpretation:

If r is close to **1**, it indicates a strong positive linear relationship.

If it is close to **-1**, it indicates a strong negative relationship.

If it is close to **0**, there is little or no linear relationship.

Part 3: Data Visualization

1. Scatterplots:

Plot scatterplots for each advertising platform (TV, Radio, and Newspaper) against Sales to visualize relationships.

Add trendlines to each plot to see if there is a linear pattern.

2. Compare Correlations:

Which advertising medium has the strongest correlation with sales? What does that mean in a business context?

Part 4: Reporting and Presentation

Report:

Include:

The Pearson correlation coefficients for each pair of variables you analyzed.

Scatterplots with trendlines for each relationship.

A written interpretation of your findings, explaining the strength and direction of each correlation.

Deliverables:

Report Submission:

Detailed report with calculations, visualizations, and interpretations.

Grading Criteria:

Correlation Calculation (20%)

Visualizations (Scatterplots, Trendlines) (20%)

Interpretation and Insights (30%)

Report (30%)